Lathi 20题练习册 第4批

对应 Boulet 前四章

原书英文题干、公式与必要图示 · 无答案版

记号：D = d/dt；差分式中 E 为前移算子，E{y[n]} = y[n+1]；j^2 = -1
u 为单位阶跃，δ 为单位冲激；LTIC / LTID 分别指连续 / 离散时间 LTI 系统

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来源：B. P. Lathi / R. A. Green, Linear Systems and Signals, 3rd ed. (2018)
中文主题仅用于课程对应；题目要求以原书英文为准。

由变换后信号反求原信号并作图

Lathi 1.2-5 | 原书第 139 页

Boulet 第1章：Transformations of the Time Variable, p.4

1.2-5 Define $2x(-3t+1) = t[u(-t-1) - u(-t+1)]$, where $u(t)$ is the unit step function.

(a) Plot $2x(-3t+1)$ over a suitable range of t.

(b) Plot $x(t)$ over a suitable range of t.

演算区

利用单位阶跃构造分段信号

Lathi 1.4-1 | 原书第 141 页

Boulet 第1章：Generalized Functions，p.26

1.4-1 Sketch the following signals:

(a) $u(t-5)-u(t-7)$

(b) $u(t-5)+u(t-7)$

(c) $t^2[u(t-1)-u(t-2)]$

(d) $(t-4)[u(t-2)-u(t-4)]$

演算区

用广义函数定义证明冲激缩放性质

Lathi 1.4-11 | 原书第 142、87 页

Boulet 第1章：Generalized Functions，p.26

所引用的定义式(1.11)已附，取 T=0；测试函数在冲激所在位置连续。缩放参数 a 非零。

1.4-11 Using the generalized function definition of impulse [Eq. (1.11) with $T = 0$], show that

$$\delta(at) = \frac{1}{|a|}\delta(t)$$

$$\int_{-\infty}^{\infty} \phi(t)\delta(t-T)\,dt = \phi(T)\int_{-\infty}^{\infty} \delta(t)\,dt = \phi(T) \tag{1.11}$$

演算区

图示信号的时间变换、能量与奇偶分解

Lathi 1.5-7 | 原书第 143 页

Boulet 第1章：Transformations of the Time Variable，p.4；Finite-Energy and Finite-Power Signals，p.21；Even and Odd Signals，p.23

本题所需原图 P1.5-7 已包含在题干中。

1.5-7 Consider the signal $x(t)$ shown in Fig. P1.5-7.

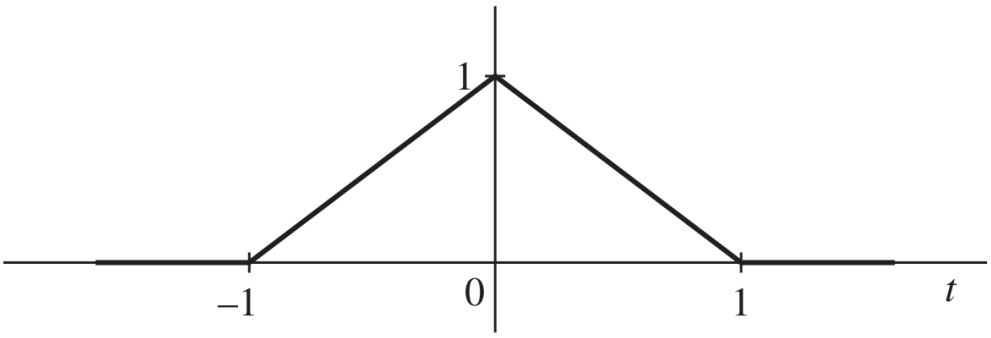


Figure P1.5-7

(a) Determine and carefully sketch $v(t) = 3x(-(1/2)(t+1))$.
(b) Determine the energy and power of $v(t)$.
(c) Determine and carefully sketch the even portion of $v(t)$, $v_e(t)$.
(d) Let $a = 2$ and $b = 3$; sketch $v(at+b)$, $v(at)+b$, $av(t+b)$, and $av(t)+b$.
(e) Let $a = -3$ and $b = -2$; sketch $v(at+b)$, $v(at)+b$, $av(t+b)$, and $av(t)+b$.

演算区

离散有限序列的能量、抽取与扩展

Lathi 3.2-4 | 原书第 316 页

Boulet 第1章：Transformations of the Time Variable，p.4；Finite-Energy and Finite-Power Signals，p.21

依照Lathi第3.2节的离散时间扩展约定，扩展后下标非整数的位置补零；向下箭头标出原序列的 n=0。

3.2-4 Letting $\downarrow$ identify $n = 0$, define the nonzero values of signal $x[n]$ as $[-1, \underset{\downarrow}{2}, -3, 4, -5, 4, -3, 2, -1]$.

(a) Determine the energy E_x and power P_x of the signal $x[n]$.

(b) Using vector form, represent signal $y[n] = x[2(n+2)]$. Be sure to identify the $n = 0$ element.

(c) Using vector form, represent signal $z[n] = x[-\frac{n-6}{3}]$. Be sure to identify the $n = 0$ element.

演算区

辨认冲激响应时移并作图解卷积

Lathi 2.4-5 | 原书第 226 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，p.67；Properties of Linear Time-Invariant Systems，p.74

原图画的是 h(t+1)，不是 h(t)；题干及原图 P2.4-5 已完整保留。

2.4-5 Suppose an LTIC system has impulse response $h(t)$ and input $x(t) = u(t)$. Figure P2.4-5 shows $x(t)$ and $h(t + 1)$, respectively. Be careful! Figure P2.4-5 shows $h(t + 1)$, not $h(t)$.

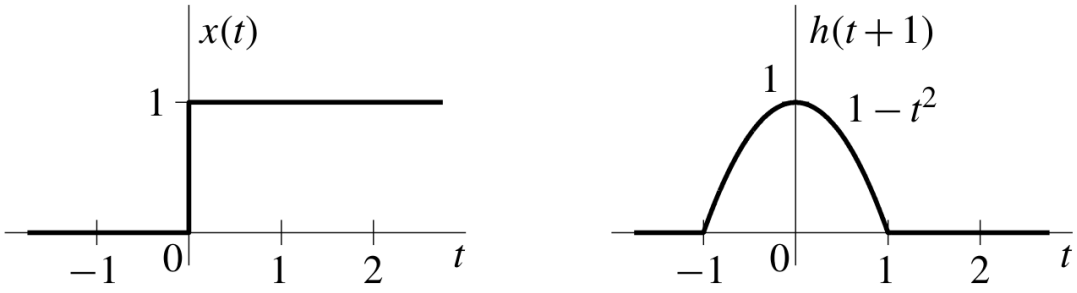


Figure P2.4-5

(a) Is system $h(t)$ causal? Mathematically justify your answer.

(b) Use the graphical convolution procedure to determine $y_{zsr}(t) = x(t) * h(t)$. Accurately sketch $y_{zsr}(t)$. When solving for $y_{zsr}(t)$, flip and shift $x(t)$ and explicitly show all integration steps.

演算区

反因果阶跃输入的两种图解卷积

Lathi 2.4-8 | 原书第 226、227 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，p.67

所需原图 P2.4-8 位于下一原书页，已附。

2.4-8 An LTIC system has impulse response $h(t)$, as shown in Fig. P2.4-8. Let t have units of seconds. Let the input be $x(t) = u(-t-2)$ and designate the output as $y_{zsr}(t) = x(t) * h(t)$.

(a) Use the graphical convolution procedure where $h(t)$ is flipped and shifted to determine $y_{zsr}(t)$. Accurately plot your result.

(b) Use the graphical convolution procedure where $x(t)$ is flipped and shifted to determine $y_{zsr}(t)$. Accurately plot your result.

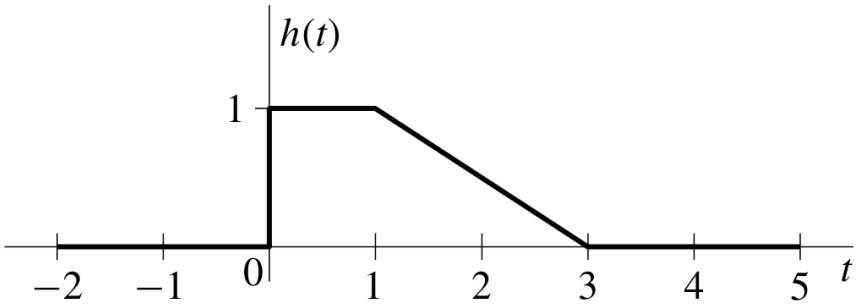


Figure P2.4-8

演算区

有限正弦脉冲与阶跃信号的卷积

Lathi 2.4-25 | 原书第 228 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，p.67

所需原图 P2.4-25 的 x(t) 和 g(t) 均已附。

2.4-25 Figure P2.4-25 shows $x(t)$ and $g(t)$. Find and sketch $c(t) = x(t) * g(t)$.

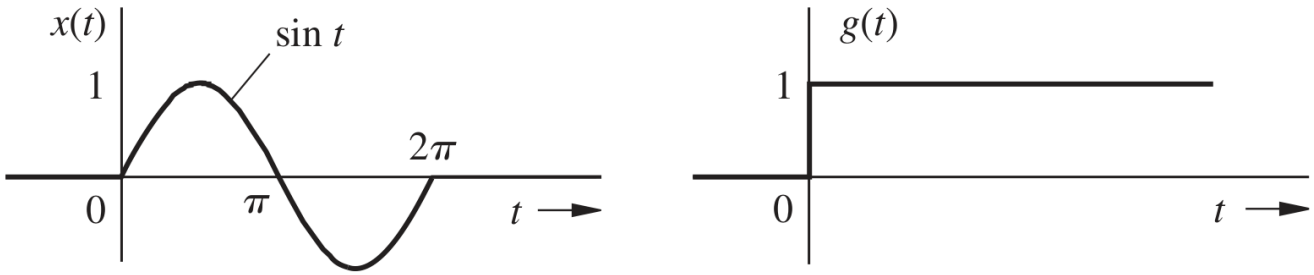


Figure P2.4-25

演算区

离散指数卷积的分段表达与临界点

Lathi 3.8-1 | 原书第 321 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，p.54

3.8-1 The convolution $y[n] = \left(\frac{5}{2^n}u[n+5]\right) * (3^n u[-n-2])$ can be represented as

$$y[n] = \begin{cases} C_1(\gamma_1)^n & n < N \\ C_2(\gamma_2)^n & n \geq N \end{cases}$$

Using the graphical convolution procedure, determine constants C_1, C_2, γ_1, γ_2, and N.

演算区

阶跃、时移与双边指数的离散卷积

Lathi 3.8-2 | 原书第 321 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，p.54

3.8-2 Use the graphical convolution procedure to determine the following:

(a) $y_a[n] = u[n] * (u[n-5] - u[n-9] + (0.5)^{(n-8)}u[n-9])$

(b) $y_b[n] = (\frac{1}{2})^{|n|} * u[-n+5]$

演算区

由微分系统求阶数、特征根与零输入响应

Lathi 2.2-2 | 原书第 223、224 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92；Characteristic Polynomials and Stability，p.112

2.2-2 Consider a linear time-invariant system with input $x(t)$ and output $y(t)$ that is described by the differential equation

$$(D+1)(D^2-1)\{y(t)\} = (D^5-1)\{x(t)\}$$

Furthermore, assume $y(0) = \dot{y}(0) = \ddot{y}(0) = 1$.

(a) What is the order of this system?
(b) What are the characteristic roots of this system?
(c) Determine the zero-input response $y_{\text{zir}}(t)$. Simplify your answer.

演算区

二阶微分系统的单位冲激响应

Lathi 2.3-3 | 原书第 225 页

Boulet 第3章：Impulse Response of a Differential LTI System，p.101

按原题引用的2.3-2要求，求给定系统的单位冲激响应 h(t)。

2.3-3 Repeat Prob. 2.3-2 for

$$(D^2+5D+6)y(t)=(D^2+7D+11)x(t)$$

演算区

二阶差分系统的带输入递推

Lathi 3.5-5 | 原书第 320 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations, p.96

3.5-5 Solve the following equation recursively (first three terms only):

$$y[n+2]+3y[n+1]+2y[n]=$$
$$x[n+2]+3x[n+1]+3x[n]$$

with $x[n]=(3)^n u[n]$, $y[-1]=3$, and $y[-2]=2$

演算区

具有振荡模态的离散零输入响应

Lathi 3.6-4 | 原书第 320 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations， p.96

3.6-4 Solve

$$y[n+2]-2y[n+1]+2y[n]=0$$

if $y[-1]=1$ and $y[-2]=0$.

演算区

离散冲激响应的闭式求解与递推校验

Lathi 3.7-2 | 原书第 320、321 页

Boulet 第3章：Impulse Response of a Difference LTI System， p.109

本题原书跨页，(a)-(h)八个小题均已附；每个系统还需递推核验 n=3 的冲激响应值。

3.7-2 Determine the unit impulse response $h[n]$ of the following systems. In each case, use recursion to verify the $n = 3$ value of the closed-form expression of $h[n]$.

(a) $(E^2 + 1)\{y[n]\} = (E + 0.5)\{x[n]\}$

(b) $y[n] - y[n-1] + 0.25y[n-2] = x[n]$

(c) $y[n] - \frac{1}{6}y[n-1] - \frac{1}{6}y[n-2] = \frac{1}{3}x[n-2]$

(d) $y[n] + \frac{1}{6}y[n-1] - \frac{1}{6}y[n-2] = \frac{1}{3}x[n]$

(e) $y[n] + \frac{1}{4}y[n-2] = x[n]$

(f) $(E^2 - \frac{4}{9})\{y[n]\} = (E^2 + 1)\{x[n]\}$

(g) $(E^2 - \frac{1}{4})(E + \frac{1}{2})\{y[n]\} = E^3\{x[n]\}$

(h) $(E - \frac{1}{2})^2\{y[n]\} = x[n]$

演算区

多谐波三角级数与复指数频谱

Lathi 6.3-4 | 原书第 672 页

Boulet 第4章：Determination of the Fourier Series Representation，p.134；Graph of the Fourier Series Coefficients: The Line Spectrum，p.137

6.3-4 The trigonometric Fourier series of a certain periodic signal is given by

$$x(t) = 3 + \sqrt{3}\cos 2t + \sin 2t + \sin 3t - \frac{1}{2}\cos\left(5t + \frac{\pi}{3}\right)$$

(a) Sketch the trigonometric Fourier spectra.
(b) By inspection of spectra in part (a), sketch the exponential Fourier series spectra.
(c) By inspection of spectra in part (b), write the exponential Fourier series for $x(t)$.
(d) Show that the series found in part (c) is equivalent to the trigonometric series for $x(t)$.

演算区

由三角幅相谱重建信号并转为复指数谱

Lathi 6.3-6 | 原书第 673 页

Boulet 第4章：Graph of the Fourier Series Coefficients: The Line Spectrum， p.137

所需原图 P6.3-6 的幅度谱、相位谱均已附。

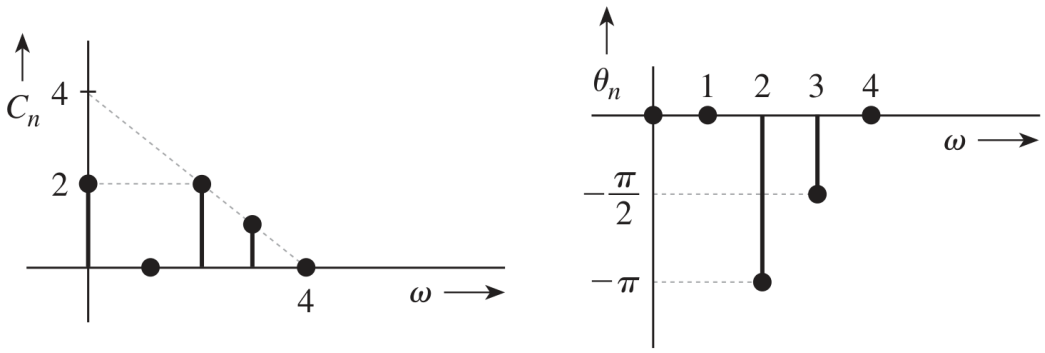
6.3-6 Figure P6.3-6 shows the trigonometric Fourier spectra of a periodic signal $x(t)$.

(a) By inspection of Fig. P6.3-6, find the trigonometric Fourier series representing $x(t)$.

(b) By inspection of Fig. P6.3-6, sketch the exponential Fourier spectra of $x(t)$.

(c) By inspection of the exponential Fourier spectra obtained in part (b), find the exponential Fourier series for $x(t)$.

(d) Show that the series found in parts (a) and (c) are equivalent.



(a) (b)

Figure P6.3-6

演算区

分段绝对值周期信号的傅里叶级数

Lathi 6.3-16 | 原书第 675 页

Boulet 第4章：Determination of the Fourier Series Representation，p.134；Properties of Continuous-Time Fourier Series，p.139

本题同页跨栏，(a)-(d)均已附。对例6.11的引用是方法提示：用傅里叶级数性质求非直流系数，而不直接积分；信号条件已完整给出。

6.3-16 A 3-periodic signal $x(t)$ is defined as

$$x(t) = \begin{cases} |t| & -1 \le t \le 1 \\ 0 & 1 < |t| \le 1.5 \\ x(t+3) & \forall t \end{cases}$$

(a) Plot $x(t)$ over $-3 \le t \le 3$.

(b) Determine D_0, the dc content of $x(t)$.

(c) Similar to Ex. 6.11, use properties and not integration to determine D_n for $n \neq 0$.

(d) Plot the magnitude spectrum $|D_n|$ over a suitable range of n. What is the most dominant frequency component of this signal?

演算区

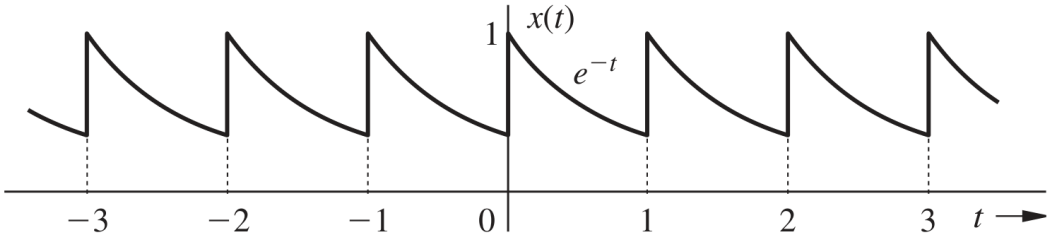
周期指数信号经过RC系统的响应

Lathi 6.4-5 | 原书第 676 页

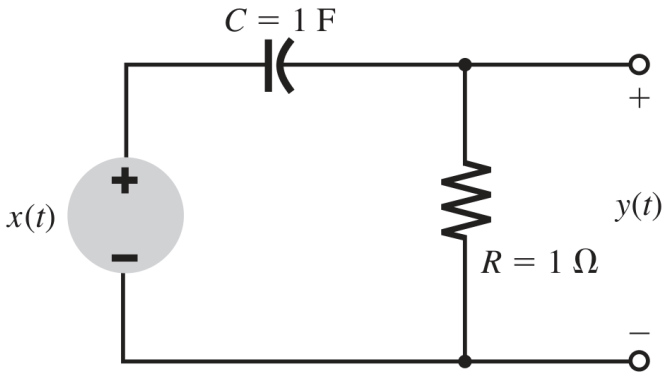
Boulet 第4章：Steady-State Response of an LTI System to a Periodic Signal，p.155

所需原图 P6.4-5(a)的周期输入和(b)的RC电路均已附，图中 C=1 F、R=1 ohm。

6.4-5 (a) Find the exponential Fourier series for a periodic signal $x(t)$ shown in Fig. P6.4-5a.
(b) The signal $x(t)$ is applied at the input of an LTIC system shown in Fig. P6.4-5b. Find the expression for the output $y(t)$.



(a)



(b)

Figure P6.4-5

演算区

正交信号的能量可加性与向量类比

Lathi 6.5-3 | 原书第 677 页

Boulet 第4章：Optimality and Convergence of the Fourier Series，p.144；Parseval Theorem，p.150

6.5-3 If $x(t)$ and $y(t)$ are orthogonal, then show that the energy of the signal $x(t)+y(t)$ is identical to the energy of the signal $x(t)-y(t)$ and is given by E_x+E_y. Explain this result by using the vector analogy. In general, show that for orthogonal signals $x(t)$ and $y(t)$ and for any pair of arbitrary real constants c_1 and c_2, the energies of $c_1x(t)+c_2y(t)$ and $c_1x(t)-c_2y(t)$ are both given by $c_1^2E_x+c_2^2E_y$.

演算区